

From operational data to clear accountability.

How AIB Life turns data sources into measurable service levels, traceable exceptions, and a monthly governance story.

SOURCE-BASED GUIDE / 1 OCTOBER 2026 / CURRENT IMPLEMENTATION

5 data source slots in one current set	5 + 3 headline SLAs + unit-linked steps	6 screens and panels for exploration	1 current governance pack per month
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THE APPLICATION IN THREE MOVES

01 **Bring the evidence**
Correspondence · cancellations · withdrawals · workflows



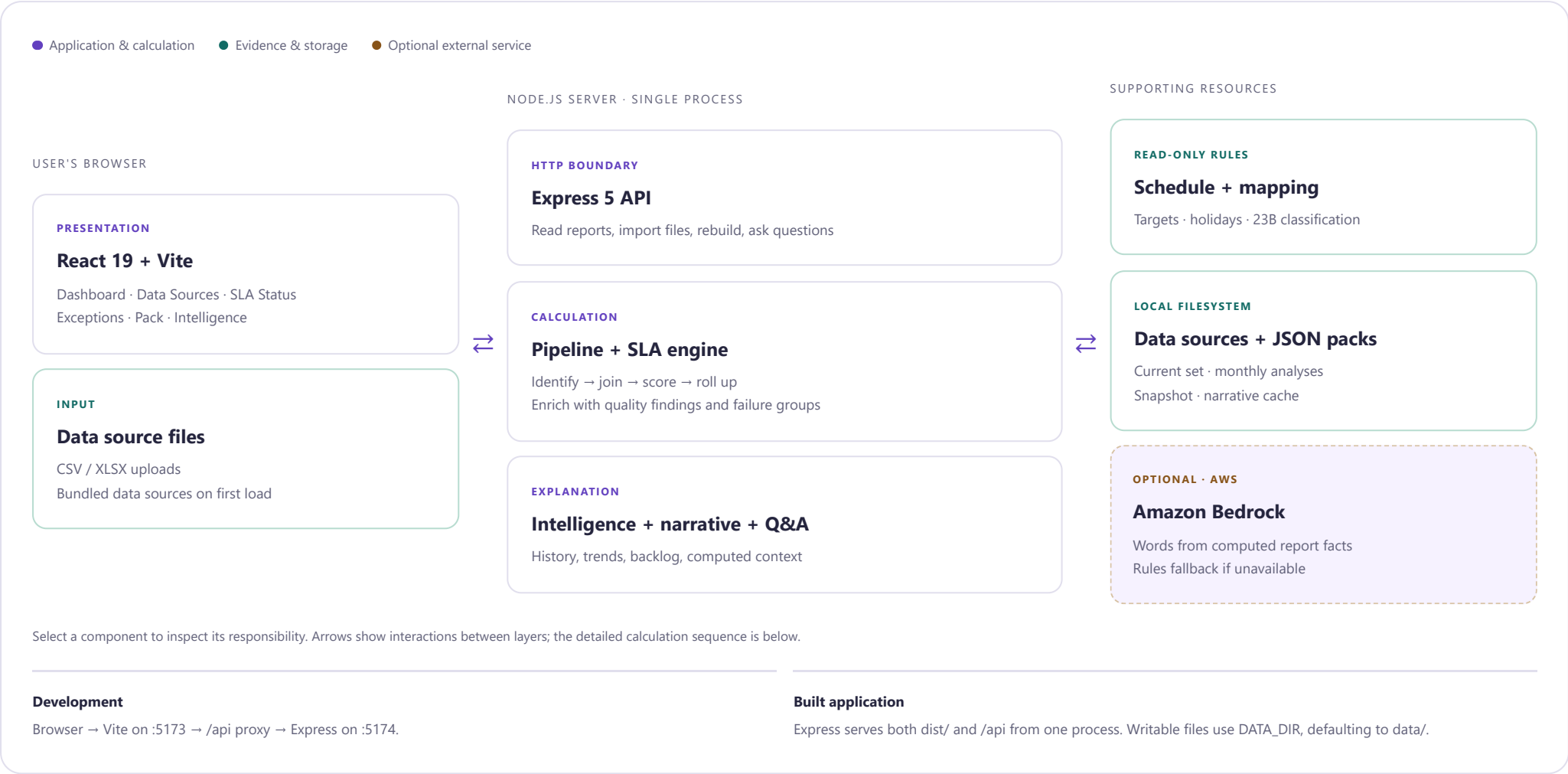
02 **Apply the agreed rules**
Business calendars · deadlines · policy outcomes



03 **Understand the position**
Monthly packs · exceptions · trends · explanation

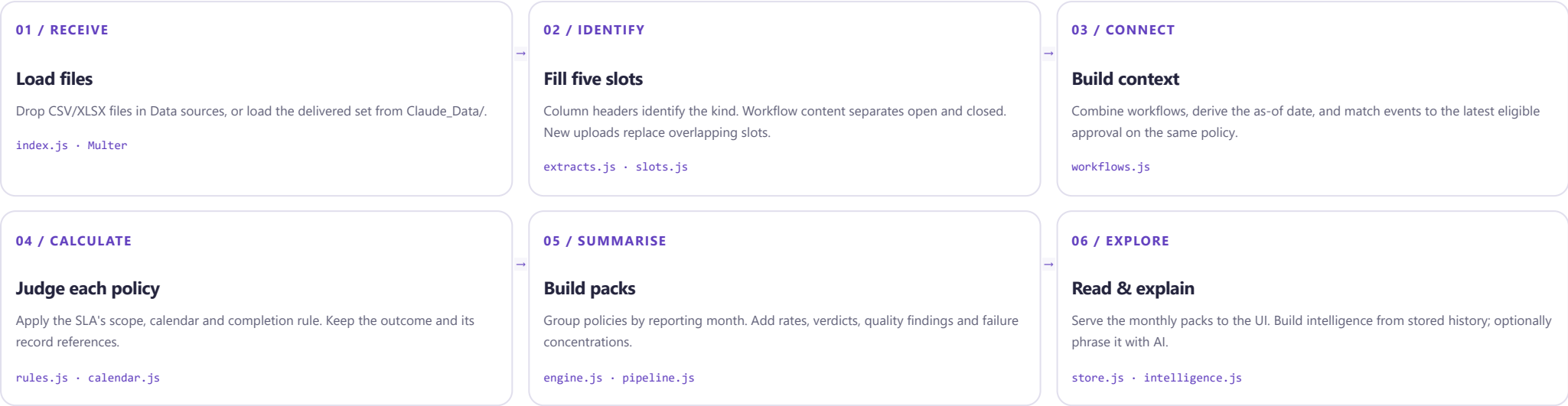
One application. Clear layers.

A React interface talks to an Express API. The server owns calculation, file storage, and the optional AI connection.



One set in. Every month out.

Data sources are a snapshot spanning many months. Replacing a file triggers a rebuild of the entire reporting history represented in that set.



The data source date anchors overdue work

Open workflow creation date + Pending Since Days gives candidate dates; the most common wins. If unavailable, the code falls back to the day after the latest activity. "Overdue" means the deadline day is before this as-of date.

Quality travels alongside the result

Set-wide findings and month-specific findings are attached to the packs. Missing files, unmatched approvals, mapping quirks and partial periods remain visible when the user interprets the numbers.

EVIDENCE SUPPLIED	FEEDS	HOW IT IS USED
EBQ correspondence	23B NUL · 23B UL Step 1	Product + transaction mapping; receipt to merged date.
CANREVEXT cancellations	23C	Protection cancellations joined to Approve Cancellation workflows.
WITHDRAWALEXT withdrawals	23B UL Step 2	Withdrawal status joined to its policy's approval workflow.
WRKFLWEXT open + closed	23A · UL Steps 2–3 · 23C · 23E	Combined workflow population supplies clocks, completions and matching context.

Reporting month follows EBQ receipt or workflow creation. Unmatched withdrawal/cancellation events use their last-status month. Uploading CSV and XLSX versions into the same slots replaces the earlier files; they are not added together.

Three ways to use the application.

These personas explain the workflow. They are not separate login roles or permission levels in the current application.

Operations analyst

"I want the report to reflect the latest available data sources."

Outcome: refreshed reporting evidence

01 / ARRIVE

Open Data sources

Use the dashboard or navigation rail. Review which of the five slots are present.

See the current evidence set

02 / SUPPLY

Drop the files

Upload CSV or XLSX data sources. Review accepted files, replacements and any skipped-file messages.

Import recognises the contents

03 / CHECK

Review completeness

Check slot coverage, the derived data source date and the set-wide quality findings.

Know what the report can measure

04 / CONTINUE

Open a period

Return to Dashboard after the automatic rebuild. Select the month to inspect its results.

Every affected month is refreshed

Recovery controls: remove a data source, rebuild the current set, or reload delivered data sources. Reloading delivered files supersedes uploads in the same slots.

Service performance reviewer

"Which service levels missed target, and which records explain it?"

Outcome: evidence for investigation

01 / ORIENT

Select a month

Scan year-grouped dashboard cards for failing SLAs, measured policies and overdue work.

Notice incomplete-period labels

02 / COMPARE

Read SLA Status

Compare completed rate with target. Inspect the three unit-linked steps and the rate including overdue work.

Separate verdict from backlog

03 / TRACE

Drill into policies

Select an SLA line to inspect record references, policy, handler, clock, deadline and completion.

Connect totals to evidence

04 / FOCUS

Filter exceptions

Use SLA and missed/overdue filters. Review failure concentrations and the month's quality findings.

Find the work to investigate

A passing SLA can still contain missed or overdue policies. The Exceptions view includes those failed policies even when the headline percentage meets target.

Governance lead

“What should the meeting understand, and what evidence can I share?”

Outcome: an explainable governance pack

01 / FRAME

Choose the period

Open a reporting month and review its position and material findings.

Start with the measured facts

02 / CONTEXT

Open Intelligence

Choose all history or a cutoff month. Review trends, pass/fail history, concentrations and backlog.

Understand the wider pattern

03 / EXPLAIN

Read & ask

Read the executive narrative or ask a question about the computed report, such as where failures concentrate.

Use the report's existing facts

04 / SHARE

Export the pack

Return to Governance pack. Select Export as PDF and choose Save as PDF in the browser print dialog.

Share a monthly document

The pack's executive summary is composed from monthly figures. The optional AI narrative appears in Intelligence. A cutoff month filters the current stored history; it does not restore a previously published snapshot.

What each screen does for you.

Six complementary views take the user from overview to evidence and then to a shareable report.

01

Governance dashboard

Groups reporting periods by year and summarises SLA performance, measured work and overdue open policies. Period cards open the monthly views.

Helps you decide where to look first.

▼ Screen & implementation

Partial periods are marked so an unfinished month is not read as a final result.

```
#dashboard · src/views/Dashboard.jsx
GET /api/bootstrap
```

02

Data Sources

Accepts files, shows coverage of the five slots, and lists file details and data-quality findings. Supports removal, rebuild and reload of delivered data.

Makes the evidence set understandable.

▼ Screen & implementation

The upload limit is 12 files per request, up to 50 MB each. Unrecognised files are skipped with messages; the expected-results workbook is rejected.

```
#data-sources · src/views/Extracts.jsx
/api/extracts · /rebuild · /bundled
```

03

SLA Status

Shows targets, met/missed counts, open work, two rates, gap to target and pass/fail. Unit-linked steps sit beneath the overall service level.

Explains the verdict and its inputs.

▼ Screen & implementation

Open the Reports menu to switch between SLA Status, SLA exceptions and Governance pack. Select an SLA row for policy evidence. Drivers and quality findings appear alongside the position.

```
#YYYY-MM/status · src/views/Position.jsx
GET /api/analysis/:month · GET /api/items/:month
```

04

SLA exceptions

Lists below-target service levels and failed policies. Filter by SLA and by missed deadline or still-open overdue work.

Narrows the investigation to concrete records.

▼ Screen & implementation

Policy tables show source identifiers, policy/product, handler, timing and outcome. More records can be revealed with Show more.

```
#YYYY-MM/exceptions · src/views/Exceptions.jsx
src/components/ItemTable.jsx
```

05

Governance pack

Turns a monthly result into a print-ready document with summary, SLA table, exceptions, findings and source-file evidence.

Provides a consistent meeting document.

▼ Screen & implementation

Export uses window.print() and the browser's PDF destination. It is not a separate server-generated PDF service.

```
#YYYY-MM/pack · src/views/Pack.jsx
```

06

Operational intelligence

A right-side panel brings together monthly trends, the SLA record, failure concentrations, overdue backlog, narrative and report Q&A.

Adds historical context to the monthly view.

▼ Screen & implementation

Scope is all history or every stored month through the selected cutoff. Trends compare rates with SLA targets.

```
#intelligence or #YYYY-MM/status/intel
src/views/Intelligence.jsx · /api/intelligence
```

What “met target” actually means.

Schedule 23 defines the targets. Policy rules determine the counts. The completed-policy rate determines the service-level verdict. Policies are counted per service record; the same policy can appear more than once.

SERVICE LEVEL	TARGET	WHEN A POLICY MEETS THE RULE
23A · Policy issue	97%	Before 15:00: close the same business day. At/after 15:00: close the same or next business day.
23B NUL · Alterations	96%	Mapped non-unit-linked EBQ policy merged by the end of the next business day after receipt.
23B UL · Unit-linked	98%	Combined counts from three steps: 1 EBQ merged the same business day; 2 withdrawal completed under the approval workflow's 15:00 rule; 3 unit adjustment workflow closed under the 15:00 rule.
23C · Cancellations	96%	In-scope protection cancellation: strictly less than 48 elapsed hours from matched approval creation to cancellation last status.
23E · Unrecognised EFT	98%	Matching Manual Review workflow closed the same or next business day. Description must contain "EFT Payment Not Recognised".

Rate (completed) → PASS / FAIL

$met \div (met + missed)$

At or above target = PASS. Below target = FAIL. No completed policies = NO_DATA, with no verdict.

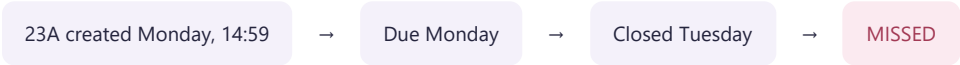
Rate including overdue open work

$met \div (met + missed + open\ overdue)$

Shows the impact of overdue unfinished work. Open policies not yet due, exclusions and unmatched policies do not enter this denominator.

ILLUSTRATIVE RECORD JOURNEY · NOT A LIVE RECORD

Why a next-day completion can still be a miss



Assuming Monday and Tuesday are business days. At exactly 15:00, the deadline shifts to Tuesday and that completion meets the rule. For 23C, 47h 59m 30s meets the rule; exactly 48h does not.

Business-day rules skip weekends and configured Irish holidays. A non-business-day start moves to the next business day, before the cutoff. The 23C rule instead compares raw elapsed hours. Rejected EBQ policies are excluded; unmatched approval events are listed without an SLA score.

Calculate first. Explain second.

Analytics are counts and ratios over evaluated policies. The optional model receives a compact brief of already-computed facts.

01 / HISTORY

Stored monthly packs

Read all months, or months through the scope cutoff. Use the current snapshot's as-of date for backlog age.



02 / ANALYTICS

Trends & concentrations

Calculate rates, pass/fail history and overdue backlog. Highlight groups with unusually high failure rates.



03 / LANGUAGE

Bedrock or rules

Use the configured model when credentials are detected. Otherwise compose the explanation deterministically.



04 / CHECKS

Accept or fall back

Reject unsupported numeric figures. The executive narrative also checks selected pass/fail and "most often" claims.

How failure concentrations work

Group measured policies by assignee, EBQ user, product, workflow or transaction type. Compare each group's failure rate with its SLA's rate.

- Minimum lift: 1.25× the SLA's failure rate.
- Monthly packs: at least 3 measured policies and 2 failures; up to 8 groups.
- Intelligence: at least 5 measured policies and 3 failures; up to 10 groups.
- Rank by excess failures. These are descriptive patterns, not demonstrated causes.

The actual AI boundary

Default model: Amazon Nova Pro through Bedrock; the code also supports an Anthropic model path. Model failures trigger a rules response.

- Narratives are cached on disk by a hash of the computed brief.
- Q&A is single-turn, report-scoped, with no retrieval or conversation memory.
- Q&A currently applies the numeric guard only. It does not invoke the narrative's claim checker.
- Checks are limited heuristics; they are not a proof that every generated sentence is correct.

Where the explanation comes from.

This guide follows the current source, including the quality module open in your editor. File paths below are relative to the project root.

Browser & navigation

`src/App.jsx` · `src/api.js`
`src/views/` · `src/components/`

Hash-based routes, API calls, screens, drill-down and print views.

API & orchestration

`server/index.js`
`server/pipeline.js` · `server/slots.js`

HTTP endpoints, serialised mutations, slot replacement and full rebuilds.

Rules & evaluation

`server/engine/*`
`config/sla-schedule.json`
`config/mapping-23b.json`

Parsing, joins, business dates, per-policy decisions and monthly roll-ups.

Data quality

`server/quality.js`

`setFindings()`: missing files and format quirks. `monthFindings()`: overdue work, unmatched records, exclusions, dependencies and incomplete periods.

Insights & language

`server/insights.js` · `server/intelligence.js`
`server/narrative.js` · `server/assistant.js`

Failure groups, historical analysis, prompts, guards, fallback and narrative caching.

Persistence & delivery

`server/store.js` · `vite.config.js`
`render.yaml` · `scripts/verify-real.js`

Filesystem storage, development proxy, hosting blueprint and expected-results comparison harness.

Current implementation boundaries

- File uploads and bundled files provide the data; there is no direct source-system connection.
- State is stored in files, with one current pack per month. Rebuilds replace prior packs; there is no versioned reporting archive.
- The API has no application login or role enforcement, and no workflow for assigning or resolving exceptions.
- Forecasts, service credits and automatic remediation are outside the implemented feature set.

Dataset & verification context

- The delivered data is documented as synthetic data in real data source formats. It represents a 24 September 2026 snapshot spanning 22 reporting months.
- The existing verification harness compares CSV and XLSX calculations with `SLA_Expected_Results.xlsx`.
- The project log records zero differences and 24 passing tests from the prior implementation verification; those engine checks were not rerun for this guide.
- Tracked follow-ups: support header-less real EBQ exports and extend the holiday list beyond 3 August 2026.

Persistence detail: imports/rebuilds are queued within the Node process. Rebuild clears and rewrites the monthly files; it is not an atomic switch of the whole dataset. Durable hosting needs a persistent `DATA_DIR`. The supplied Render blueprint documents ephemeral storage on its free configuration.